Maya Forecasting System

Day-ahead forecasts of PacifiCorp East demand, Milford, Utah renewable output and net load. Every forecast uses only information available the day before.

Evaluation period

2024-25 backtest

Demand error (MWh)

26.7% better than PACE (254.7 MWh)

186.7

Renewable error (MWh)

30.7% better than persistence (18.7 MWh)

12.9

Net load error (MWh)

27.8% better than baseline (258.9 MWh)

186.9

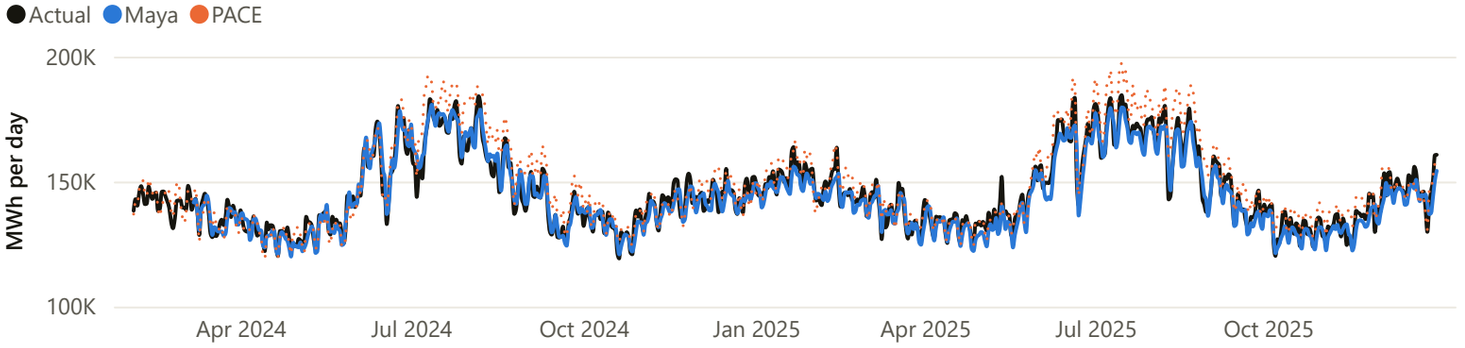
Demand MAPE

15,966 hours scored

3.0%

Daily demand: actual vs Maya vs PACE

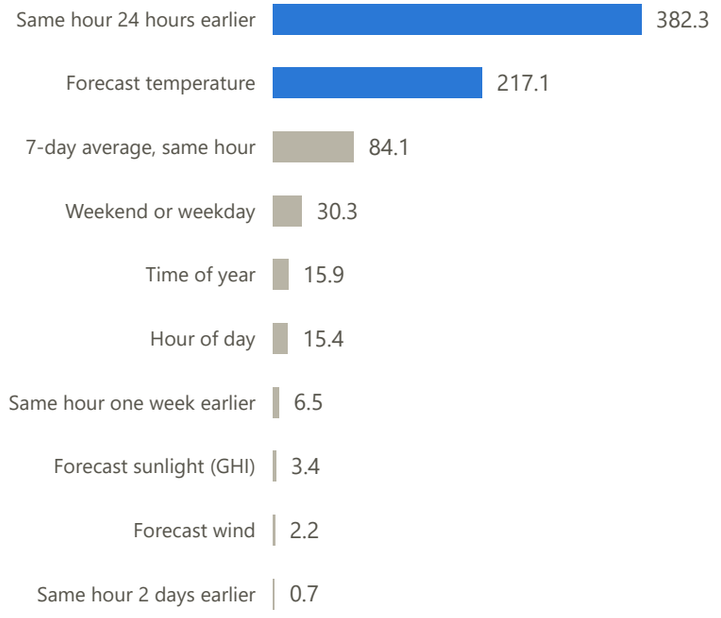
MWh per day, local (Mountain) dates. Days with missing hours are left out.



What drives the forecast

Demand

Extra error when each input is scrambled (MWh)



Model comparison

Model	Hours	MAE (MWh)	Baseline MAE	Improvement %	RMSE (MWh)	MAPE %
Demand (blend)	15,966	186.7	254.7	26.7	243.7	3.03
Demand, GBM only	15,966	199.0	254.7	21.9	259.9	3.20
Demand, LSTM only	15,966	190.3	254.7	25.3	249.6	3.11
Renewable output	15,967	12.9	18.7	30.7	26.2	5.54
Net load	15,966	186.9	258.9	27.8	244.4	3.14

How it works

Demand blends gradient boosting and an LSTM. Renewable output uses gradient boosting on forecast sunlight. Net load = demand minus Milford renewables.

Data: EIA-930, EIA-923, Copernicus ERA5 and CAMS, ECMWF (Open-Meteo), EPA eGRID.

Forecast explorer

Pick a series and a date range. Drill down on the chart (double-arrow icon) to go from days to hours.

Evaluation period

2024-25 backtest

Series

Demand

Date range (local)

1/1/2024 — 12/31/2026

Maya MAE in range (MWh)

15,966 hours scored

186.7

Baseline MAE in range (MWh)

PACE day-ahead forecast

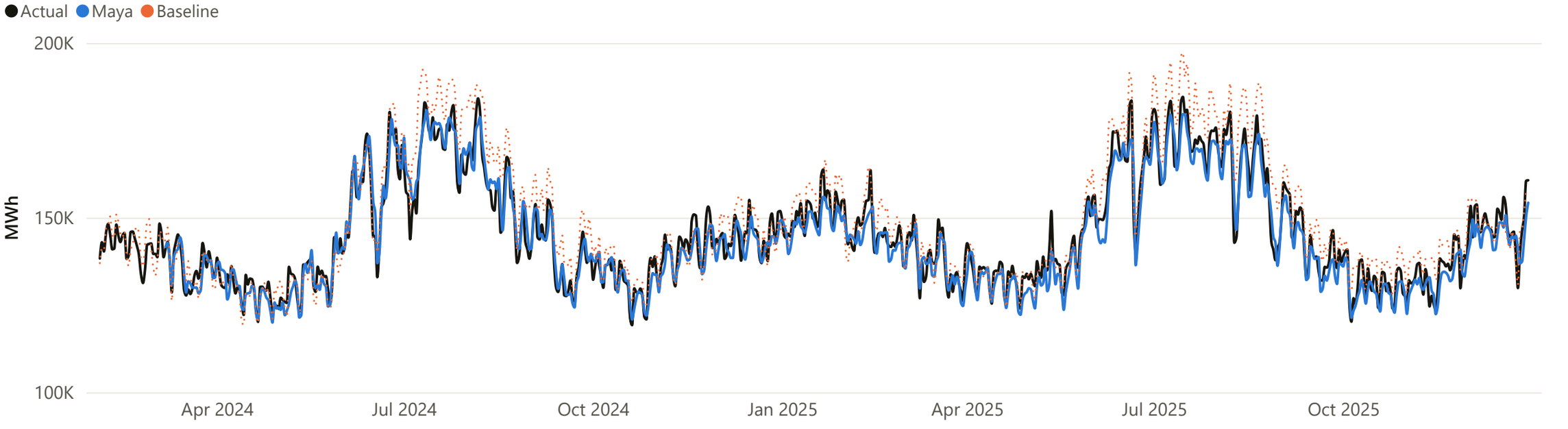
254.7

Maya improvement over baseline

26.7%

Demand: actual vs Maya vs PACE day-ahead forecast (MWh)

Daily totals by default; drill down to see hours of a day.



Where PACE misses

PACE's own day-ahead forecast runs high in summer, especially in the evening. Positive bias means the forecast was too high.

Evaluation period

2024-25 backtest

Hours PACE misses by over 500 MWh

12.4%

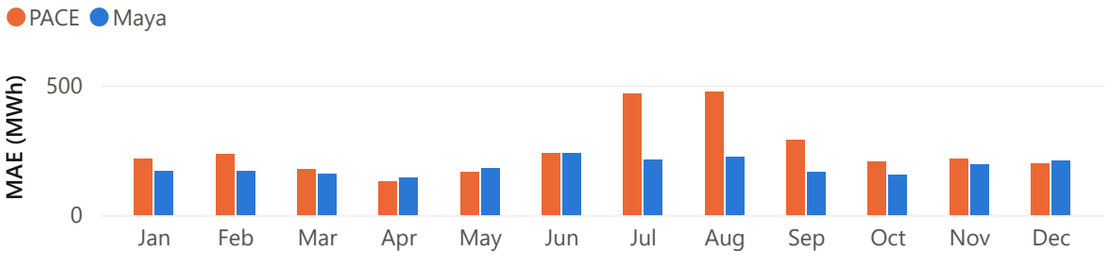
Hours Maya misses by over 500 MWh

4.4%

PACE's big misses that are over-forecasts

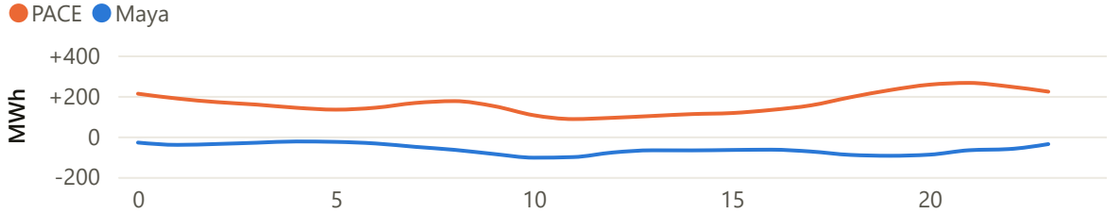
96%

Average error by month (MAE, MWh)



Average bias by hour of day (local)

Forecast minus actual, MWh. Above zero means too high.



PACE bias by month and local hour (MWh). Orange: PACE too high, blue: too low

Month	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Jan	+200	+145	+161	+165	+165	+170	+184	+200	+216	+203	+131	+85	+71	+51	+48	+19	+26	+76	+186	+254	+257	+229	+213	+214
Feb	+189	+148	+152	+145	+159	+167	+185	+220	+243	+245	+158	+66	+31	+10	-1	-7	-12	+18	+119	+175	+164	+165	+173	+168
Mar	+89	+81	+90	+75	+68	+47	+61	+84	+103	+109	+54	-4	-27	-30	-27	-24	-14	-14	-11	+20	+75	+97	+90	+88
Apr	+28	+14	-2	-6	-4	-12	+4	+28	+61	+40	-18	-46	-58	-83	-91	-93	-82	-71	-57	-18	+22	+71	+60	+47
May	-38	-24	-26	-7	-12	-15	-3	+21	+29	-12	-49	-73	-87	-94	-96	-90	-77	-81	-76	-60	-44	-14	-22	-41
Jun	+256	+230	+191	+161	+130	+126	+119	+134	+120	+73	+46	+54	+81	+107	+126	+147	+161	+176	+185	+205	+236	+282	+301	+268
Jul	+533	+497	+443	+420	+357	+336	+330	+362	+347	+285	+296	+354	+410	+457	+488	+519	+538	+536	+532	+539	+589	+632	+597	+553
Aug	+583	+558	+508	+472	+439	+398	+377	+371	+382	+335	+265	+259	+306	+364	+397	+427	+500	+543	+556	+598	+664	+670	+627	+602
Sep	+333	+305	+302	+274	+240	+199	+211	+254	+273	+257	+182	+142	+136	+150	+174	+189	+219	+257	+327	+410	+472	+443	+398	+342
Oct	+185	+157	+131	+133	+136	+130	+143	+156	+161	+165	+151	+127	+104	+90	+96	+86	+95	+109	+180	+267	+303	+280	+246	+212
Nov	+100	+78	+54	+41	+24	+48	+92	+125	+122	+101	+43	+12	+21	+29	+34	+39	+33	+94	+176	+178	+144	+131	+128	+102
Dec	+40	+2	+0	-3	+5	+18	+32	+63	+85	+65	+29	+28	+57	+72	+68	+40	+36	+64	+137	+136	+114	+92	+72	+47

Scenarios

Left: how added Milford solar would change net load. Right: the production models rerun for a 2026 day with changed weather.

Evaluation period: 2024-25 backtest

Added solar capacity (MW AC): 500

Months: Multiple selections

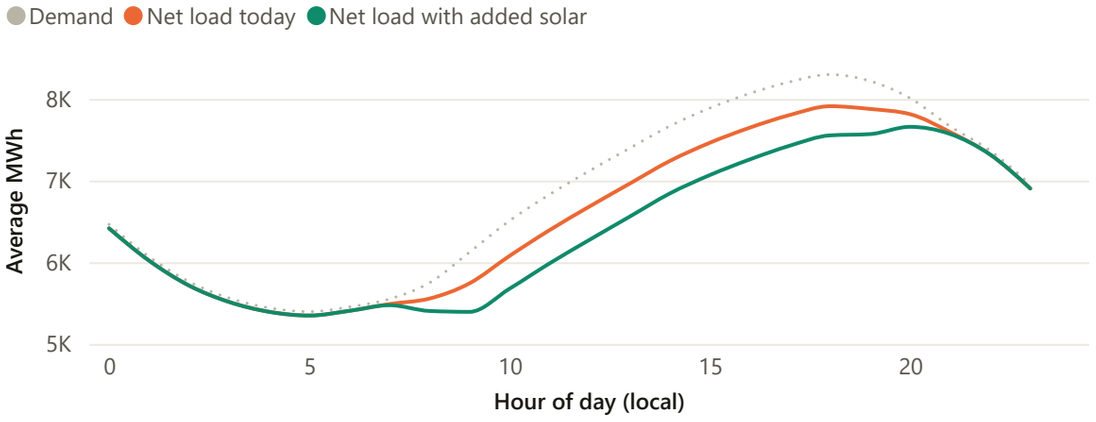
2026 day: Aug 31, 2026

Temp change (F): 0

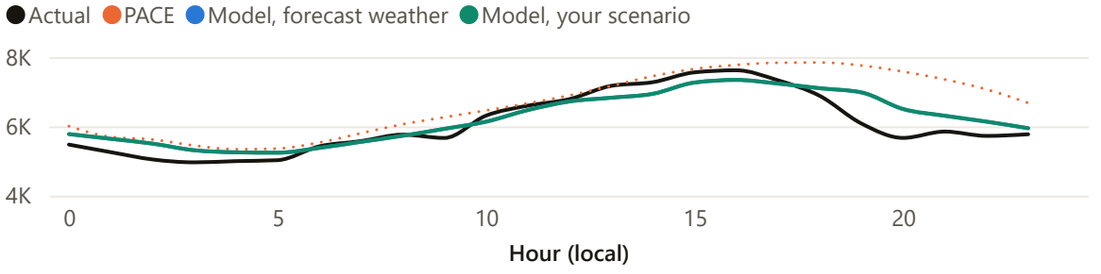
Sunshine (%): 100

Average day: net load with added solar (MWh)

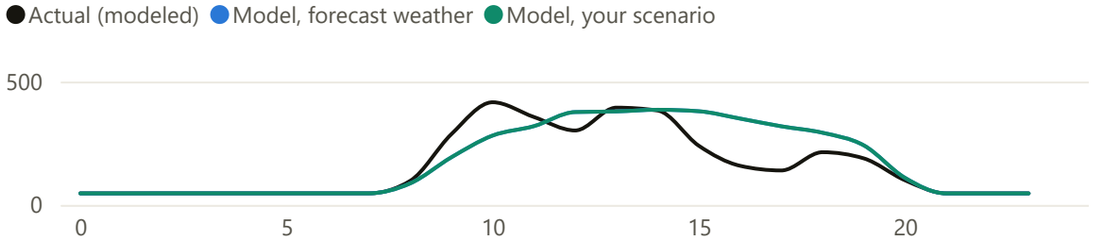
Added solar follows the hourly shape of today's 476 MW of Milford solar.



Demand for the chosen day (MWh)



Milford renewable output for the chosen day (MWh)



Average evening peak (MWh)

-252 MWh vs today

7,658

Midday low, 9 am-4 pm (MWh)

-349 MWh vs today

5,393

Midday-to-peak ramp (MWh)

+97 MWh vs today

2,265

Milford renewable share

+2.6 pts vs today

5.8%

Daily demand (MWh)

0 MWh vs forecast weather

149,224

Renewable (MWh)

0 MWh vs forecast weather

4,230

Peak net load (MWh)

0 MWh vs forecast weather

6,992